Software Engineering

Lab Work: Class Diagrams

Hiba Mhirit

October 15, 2026

Exercise 4

We want to manage flights booked through an agency. Based on interviews conducted with agency staff, we know that:

- Airlines offer different flights.
- A client can book one or multiple flights.
- A reservation concerns a single flight and a single passenger.
- A reservation can be confirmed or canceled.
- A flight has a departure airport and an arrival airport.
- A flight has a departure date and time and an arrival date and time.
- A flight may include one or more stopovers at different airports.
- A stopover has a departure time and an arrival time.
- Each airport serves one or more cities.

Class diagram

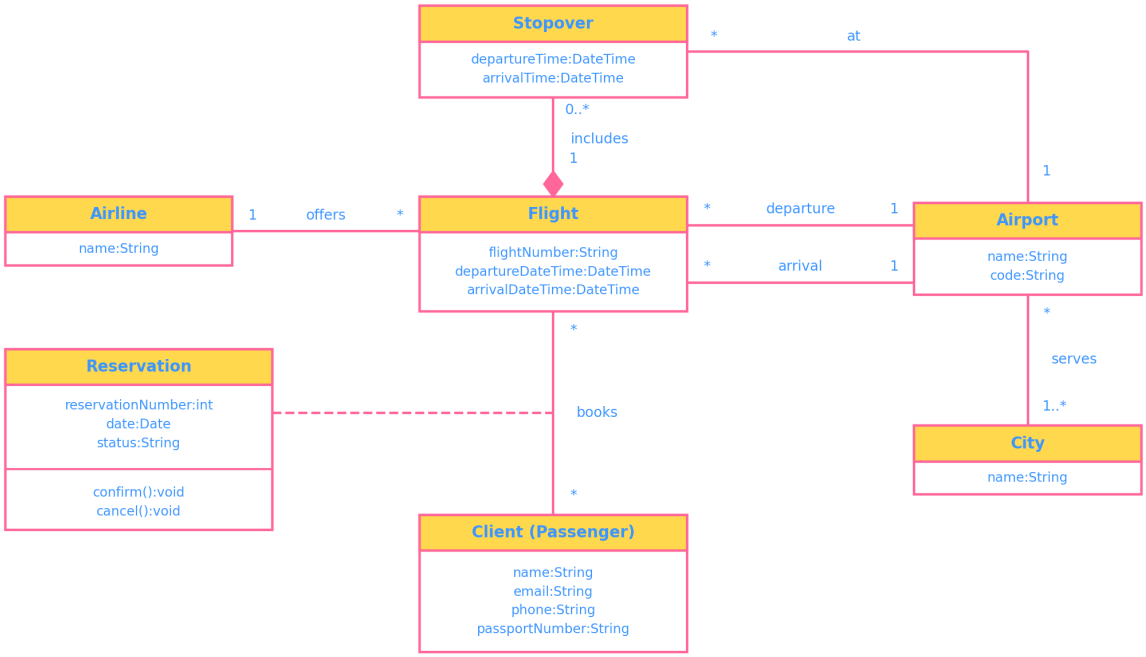


Figure 1: Class diagram of the flight booking agency.

Explanation

- A client and a passenger are the same person, so they are modelled by one class *Client (Passenger)*.
- *Reservation* is an association class between *Client* and *Flight*: a client can book several flights, a flight can be booked by several clients, and each reservation concerns one client and one flight. It has a status (confirmed or canceled) and the operations `confirm()` and `cancel()`.
- An airline offers many flights (1 – *).
- A flight has one departure airport and one arrival airport (two associations to *Airport*).
- A flight includes 0 or more stopovers (composition: a stopover cannot exist without its flight), and each stopover takes place at one airport.
- An airport serves one or more cities (1..*).